

Understanding the Physics of the Cellular Glycocalyx

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ABSTRACT

The Glycocalyx is a layer of complex polymers situated on the outer surfaces of most cells, which plays an important role in the regulation of cell surface receptors and signalling between cells. The polymers are embedded in the plasma membrane, and thermodynamically and kinetically regulate the reception of extra-cellular signals by cell-surface receptors. The lower layer of the glycocalyx consists of bottlebrush polymers, with an outer layer of linear polymers that extend further out. Physically, the glycocalyx has been modelled as a polymer brush, a system for which many scaling laws are known. However, the properties of mixed polymer brushes with polymers with different lengths are not well known. In this project, we study the physical properties of mixed bi-disperse polymer brushes, and show that while the scaling laws for homogeneous brushes still apply to mixed brushes, emergent behavior is also observed.

The mechanisms by which viruses or other external agents may enter the glycocalyx are also not well-known, although some hypotheses have been proposed. One such hypothesis is that viruses may rely on weak multivalent attraction to the polymer chains in the glycocalyx, but this model assumes a homogeneous brush and uniform attraction between the viruses and the polymer chains. We extend this model by introducing the viruses into the mixed brush systems studied in the first part of the project. This allows us to study the effects of non-uniform attraction between the viruses and the two types of chains in the brush, in addition to the effects of varying virus size and brush density.

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Ayush Gurule

DECLARATION

I hereby declare that the investigation presented in the thesis has been carried out by me. The work is original and has not been submitted earlier as a whole or in part for a degree/diploma at this or any other Institution/University.

Ayush Gurule

Contents

1	Introduction	1
1.1	Composition and structure	1
1.1.1	Communication layer	2
1.1.2	Outer structural layer	4
1.2	The Glycocalyx is dynamic	4
1.3	Biological Significance	7
1.4	Towards a physical model	8
1.4.1	The Bead-Spring Model for polymers	8
1.4.2	Flory Theory	12
1.4.3	Grafted Polymer Brushes	13
1.4.4	Alexander-de-Gennes scaling laws	13
1.4.5	Virus - Glycocalyx interactions	15
1.5	Project outline	17
2	Methodology	19
2.1	LAMMPS	19

2.1.1	Simulation setup	20
2.1.2	Workflow	22
3	Results	25
3.1	Verifying Alexander - de-Gennes scaling laws	25
3.1.1	Flory regime and brush regime	26
3.2	Studying mixed brushes	27
3.2.1	Mixed brushes follow scaling laws	27
3.2.2	Mixed brushes show unexpected behavior	29
3.2.3	Height difference increases with grafting density	30
3.3	Studying viral entry into the glycocalyx	32
3.3.1	Size dependence	33
3.3.2	Affinity dependence	35
4	Discussion	41
4.1	Restricting virus diffusion	41
4.2	Approximating surface receptors	42
4.3	Equilibrium Sampling for viruses	43
	Appendix	45
	Acknowledgements	48
	Bibliography	48

Chapter 1

Introduction

The glycocalyx is a dense layer of polymers and complex sugar chains, called glycans, which are embedded in the plasma membrane and extend outward from the cellular surface. It is found on the surfaces of most cells, and plays an important role in many cellular functions, including signaling between cells, cell adhesion, and other processes that involve mechano-sensing [1]. It also acts as a defensive layer, protecting the cell from physical and chemical injury [1]. Figure 1.1 shows an example of the glycocalyx in a bacterium.

The mammalian glycocalyx, in particular, has increasingly been studied as an organelle that plays a vital role in many cellular functions [2], but the complexity of its structure has posed a significant challenge to detailed investigation through both experimental and simulation based approaches.

1.1 Composition and structure

The glycocalyx is composed of various polymers, including linear polyelectrolytes like Hyaluronic and polysialic acids, as well as complex sugar macromolecules called glycans. Functionally, these can be divided into two distinct layers: the lower com-

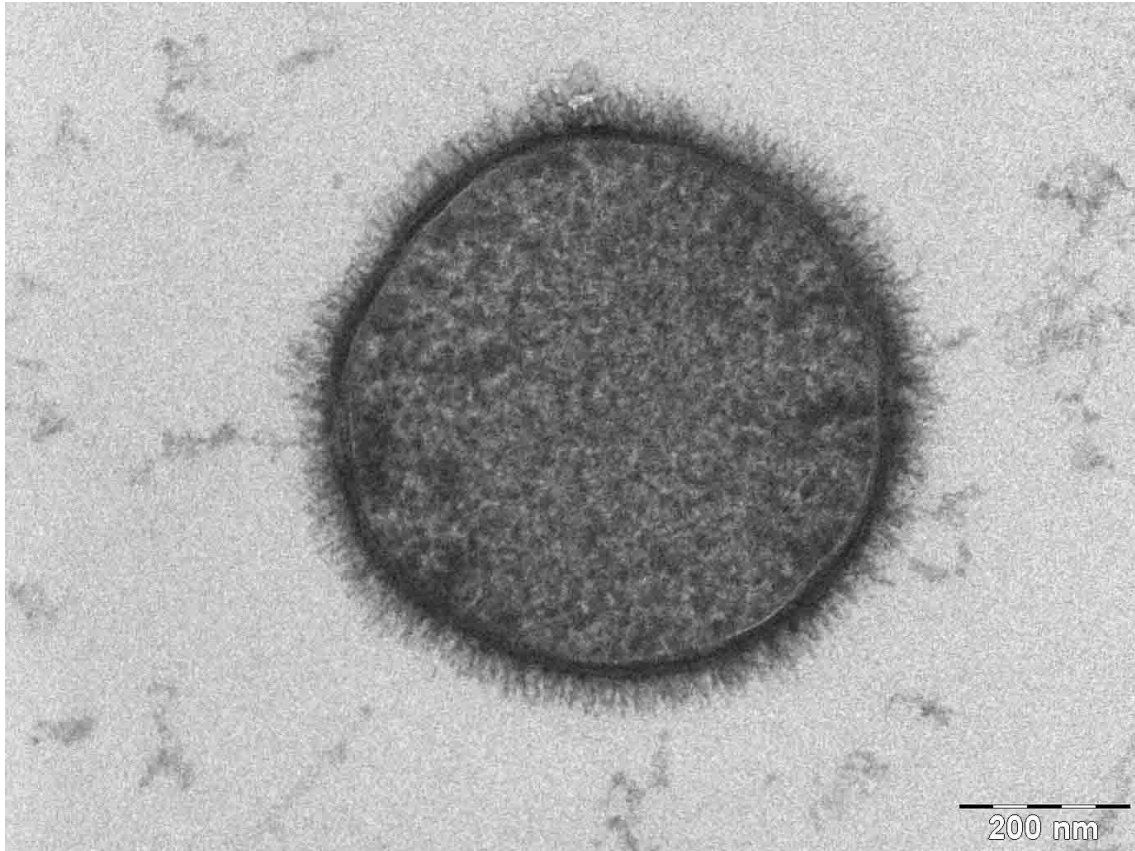


Figure 1.1: An image of the bacterium *Bacillus subtilis*, where the glycocalyx can be seen as the layer of hair-like bristles on the periphery of the cell. The size, density, and composition of the glycocalyx can vary widely depending on the cell, as well as the cell state. Image from Wikimedia Commons, licensed under CC BY-SA 3.0[3]

munication interface consisting primarily of glycans with side-chains, cell surface receptors, and other trans-membrane proteins, and the linear polymer layer consisting of long-chain polyelectrolytes which extend through and above the lower layer [1].

1.1.1 Communication layer

The communication layer, as the name suggests, plays a more important role in the regulation of the cell-surface receptors which are embedded in it. Some of the primary constituents of this layer are complex carbohydrates called glycans, or glycoproteins (they also give the glycocalyx its name - *glyco*: sugar, *calyx*: husk).

These are polysaccharides, which means they are composed of a large number of repeating saccharide monomers. The glycoproteins in the communication interface are also found to have a large number of glycan side-chains and, following common polymer nomenclature, are referred to as bottlebrush polymers. These proteins are found to be significantly shorter than the linear polyelectrolytes.

However, due to the large number of side-chains, these polymers can be massive. MUC16 is the most massive of these, with a molecular weight as high as 20 million daltons, and a length of 1500 nm when fully stretched (including the length of side-chains).

The side-chains also cause these polymers to have large persistence lengths, of the order of 10 nm [1] which is of the same order of magnitude as the length of the side-chains (5 - 10 nm) themselves [1]. The density and length of the glycan side-chains can vary widely, and cancer cells in particular are known to have shorter but more densely grafted glycan side-chains. The details of how cancer-related changes affect the mucins, which are some of the most common bottlebrush polymers in this layer, require further study.

In epithelial cells with clearly defined polarity, the mucins are usually found on the apical surface of the cell. However, in cancerous cells, there may be a loss of cell polarity, leading to the mucins being found on all sides of the cell-surface which can affect important cellular functions to drive cancer progression [1].

The bottlebrush polymers found in the communication interface are also polyelectrolytes, and can thus be charged when in solution, a fact that is exploited by cells to regulate the cell surface receptors that sit amidst them. Since the glycocalyx is usually studied in epithelial cells, which are immersed in blood flow, charge effects are all the more important for understanding the receptor regulation in the communication interface.

Due to both the density and complexity of the communication interface, creating an accurate physical model for it has proven difficult. Most studies do not distinguish between this interface and the outer layer, but the cell surface receptors are often modelled explicitly since their length is usually much smaller than the mucins and other polymers that they are embedded between.

1.1.2 Outer structural layer

Hyaluronic acid is one of the major components of the glycocalyx, and is a long-chain linear polyelectrolyte. Another similar polymer is polysialic acid, which is also found in the glycocalyx. These polymers extend through and above the communication layer, and are synthesized at the plasma membrane. They can contain over 10,000 monomers, with a total length of the order of $10\text{ }\mu\text{m}$, but are found in a random coil configuration which reduces the thickness of the layer by an order of magnitude [4]. These polymers are less stiffer than the bottlebrush polymers in the communication layer, and have persistence lengths of around 5 nm. The thickness of this layer can change significantly depending on the environmental context of the cell, and it also allows the cell to affect the functioning of a neighboring cell by interacting with its glycocalyx [1].

Cancer cells, once again, show significant alteration in this layer. The layer is found to be denser, which affects cell-cell junctions, and may also help maintain a higher pH in their peri-cellular region [1].

1.2 The Glycocalyx is dynamic

The plasma membrane, in which the transmembrane proteins of the glycocalyx are embedded, is not completely static. The lipids that constitute the bilayered

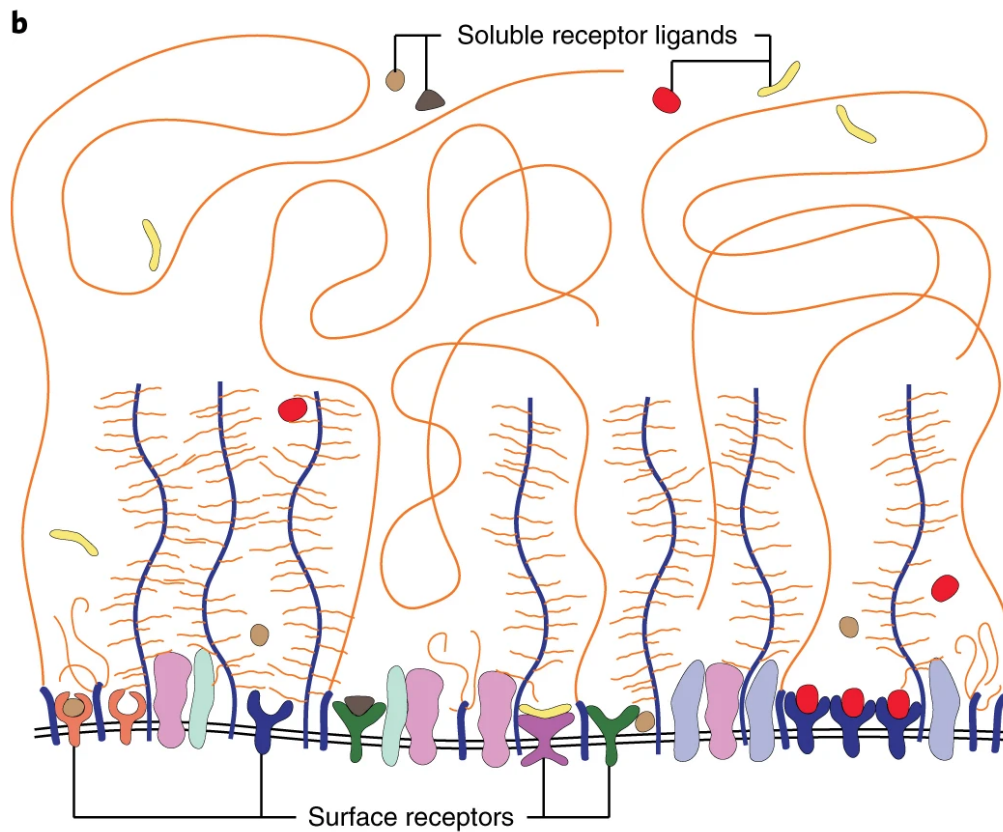


Figure 1.2: The two distinct layers in the glycocalyx can be seen here. The inner communication interface consists of various cell surface receptors, and bottle-brush polymers with numerous glycan side-chains. The outer layer consists of much longer, linear polyelectrolytes. Together, both of these layers regulate the entry of extracellular ligands through the glycocalyx to reach the surface receptors. Figure reproduced from [1]

membrane can move around, hence causing the glycocalyx proteins to also be mobile. The density and composition of both the plasma membrane and the glycocalyx are also both known to vary widely depending on what state the cell is in, which can depend on external stimuli.

Furthermore, the bilayer is composed of various different types of lipids, which show phase separation in the presence of certain factors like cholesterol. This leads to the formation of lipid rafts, where lipids of the same type gather to form large regions surrounded by different lipids. This separation can also cause the glycoproteins to be segregated, making the glycocalyx inhomogeneous and adding to the complexity of the system.

Due to its positioning on the outer surface of the cell membrane, the glycocalyx also frequently comes into contact with blood flow, especially in the endothelial cells in our body. This can also affect its structure and density, changing its permeability for various ligands, including extracellular signals. Due to this, the functioning of the glycocalyx as a cell surface receptor regulator can be significantly affected by shear flow, but such effects have not been explored significantly as of yet. Shear flow can also affect the mechanosensing function of the glycocalyx [1], which has the potential to influence the nature and functioning of the cell-cell junctions and the Extra-Cellular Matrix (ECM). The ECM also plays an important role in inter-cell communication through mechanisms that often rely on external ligands that bind with the surface receptors or those found in the ECM. The entry and binding of these external ligands may again be regulated by the glycocalyx, which has important implications for the mobility of cancer cells [1].

1.3 Biological Significance

The reception of certain extra-cellular signals by the cell-surface receptors is required for many essential cell functions, including apoptosis or programmed cell death, which is used by the body to terminate cells with defective cellular machinery. It is commonly understood that the glycocalyx regulates this signal reception, but the details of the mechanisms by which it does so are unclear due to the size and complexity of the system. Cancer cells provide important evidence regarding this function of the glycocalyx, as nearly all cancers have been shown to have a heavily modified glycocalyx layer [1]. This serves multiple purposes, one of which is to prevent apoptotic signals from the rest of the body from reaching the cancer cell, so that it may continue to proliferate. The glycocalyx can also affect cell-cell junctions, as well as junctions between the cell and the extra-cellular matrix in a tissue [1].

The glycocalyx layer is composed primarily of polyelectrolytes, which become charged when immersed in water. The thickness and density of the outer layer of the glycocalyx can prevent these charges from escaping, leading to a lowered pH level around the cell. This can serve as another way of regulating signal reception, as the cell surface receptors have evolved to work optimally at a near-neutral pH [1] and may not be able to bind to the appropriate ligands at a higher pH level. An increased pH level has also been shown to affect the bonds in the extra-cellular matrix of the cell, which can allow cancer cells to disengage from the tissue they are embedded in and become more mobile [1], a process that is essential for cancer progression.

In addition to regulating the binding of external signals or ligands to the cell-surface receptors, the glycocalyx can also regulate or, at times, outright prevent the entry of viruses and other pathogens. In order to reach the cell surface and initiate infection, most viruses need to enter through the glycocalyx, which can prove difficult, especially for larger viruses, due to the density of the layer. The dynamics at this scale cannot be studied experimentally, and the mechanisms involved in the entry

of viruses into the glycocalyx are thus not very well known.

1.4 Towards a physical model

As discussed so far, the glycocalyx is composed primarily of transmembrane proteins of various kinds. The layer is constructed, maintained and transformed through various complex chemical reaction pathways, and the interactions between the layer and any external ligands, including extracellular signals as well as viruses and other pathogens, often have a chemical basis as well. However, most of the constituent proteins of the glycocalyx can be modelled physically as polymers that interact with each other according to certain carefully tuned interaction potentials. The interaction of grafted polymers can be studied using known theories of polymer brushes, and some of the scaling laws for such systems may be considered to be applicable to the glycocalyx.

While such a model is certainly reductive, it can serve as a good starting point. The interaction potentials can be tuned to fit observed behavior, and the physical properties, such as the length, molecular architecture, persistence length, stretchability, etc. of the polymers can be modified to better represent the glycocalyx proteins.

We thus use polymer theory (described in the following sections) in order to develop such a model for the glycocalyx.

1.4.1 The Bead-Spring Model for polymers

The bead-spring model is one of the simplest ways to model a polymer. As the name suggests, it involves modeling a polymer as a collection of beads that have a finite volume and are attached to each other through bonds that can be represented by a harmonic oscillator or spring. The interactions between the beads may

be represented by a simple excluded volume or hard-core potential, which is zero everywhere and infinite when there is an overlap, or by more complicated potentials that are more realistic. A common choice is the Lennard-Jones 12-6 potential [5] (Eq. 1.1), where ϵ , the depth of the potential well (the minimum value of V), and σ , the distance between two particles at which the potential goes from negative to positive (i.e. attractive to repulsive), are the two parameters which can be varied.

$$v_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (1.1)$$

A further simplification can be made by ignoring the attractive part of the potential, but still using a smooth potential rather than a discontinuous one like the hard-core potential. This involves using a shifted Lennard-Jones potential, and introducing a cut-off distance beyond which the potential falls off to zero. If we choose the cut-off to be $2^{1/6}\sigma$, the point where the Lennard-Jones potential is minimum, we are left with only the repulsive part after adding a shift of $+\epsilon$. This shifted potential, known as the Weeks-Chandler-Andersen potential [6], is given by Eq. 1.2. Both of these interaction potentials are shown in Figure 1.3.

$$V_{\text{WCA}} = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] + \epsilon & \text{if } r \leq r_c \\ 0 & \text{if } r > r_c \end{cases} \quad (1.2)$$

The bonds between two successive monomers can be modeled as simple harmonic oscillators, with the parameter k representing the stiffness of the bond, and r_0 representing the equilibrium bond-length at which neither of the particles exerts a force on the other. The total potential from all of the bonds in a polymer is given by Eq. 1.3, where r_i represents a given bond. A polymer with N monomers has $N - 1$

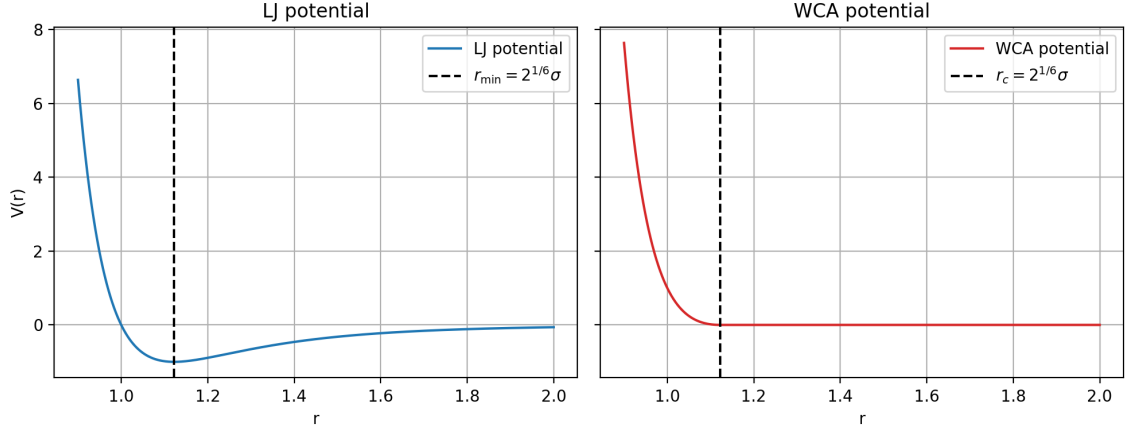


Figure 1.3: The Weeks-Chandler-Andersen potential is the Lennard-Jones potential shifted to make it purely repulsive. A cutoff is also introduced to make simulations computationally efficient.

bonds.

$$V_{\text{harmonic}} = \sum_{r_i} k(r_i - r_0)^2 \quad (1.3)$$

However, such bonds can be extended indefinitely, although the energy required to do so increases quadratically. Not only can such indefinite extension complicate the dynamics of the system, it is also not physical as approximating complex proteins or macromolecules as polymers requires the bonds to not extend significantly.

Thus, harmonic bonds are often replaced with Finite Extensible Non-linear Elastic (FENE) bonds (eq. 1.4). We see that the first term in this expression prevents the bonds from extending significantly beyond the maximum length (R_0). The second term is simply the Lennard-Jones potential, which ensures that the monomers connected by this bond cannot overlap. The lengths of such bonds are thus constrained between a finite range.

$$E = -0.5KR_0^2 \ln \left[1 - \left(\frac{r}{R_0} \right)^2 \right] + 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] + \epsilon \quad (1.4)$$

Finally, we also take into account the flexibility of the polymers via a bending potential, with the parameters κ and θ_0 controlling the stiffness and the equilibrium

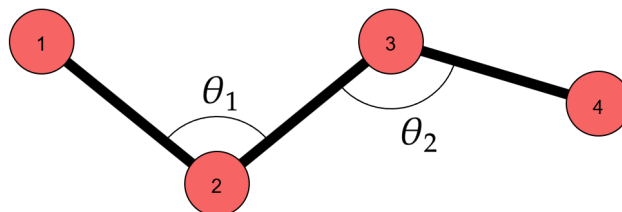


Figure 1.4: An N -bead polymer has $N-2$ angles, labeled starting from the first monomer, as shown here.

angle between three successive monomers respectively. Figure 1.4 shows the way the angles are defined in two dimensions, which can be similarly generalised to three dimensions. A κ of zero represents freely jointed chains, where the inter-monomer bonds may be stiff, but the chain is completely free to rotate along all bonds. On the other hand, a κ of ∞ represents a completely rigid polymer, which can only rotate as a whole. Such a bending potential can be designed to be harmonic in θ , but it is found to be more convenient to make it harmonic in $\cos \theta$. The latter (shown in Eq. 1.5) reduces to the former when the bending is small, and does not pose the same problems at $\theta_0 = \pi$, which is a commonly occurring scenario. The equilibrium angle is determined by the chemical composition of the polymers, and a complex heteropolymer may have different types of bonds, each with a different equilibrium angle.

$$V_B = \sum_{\theta_i} \kappa (\cos \theta_i - \cos \theta_0)^2 \quad (1.5)$$

It has been found that the flexibility of polymers affects various physical properties. For example, it is found that stiffer polymers (higher κ) adsorb onto flat, homogeneous surfaces better than more flexible ones, likely due to a lower loss in conformational entropy upon adsorption [7]. In the context of the glycocalyx, almost none of the proteins that make up the layer can be considered to be completely flexible polymers, and it is thus essential to take into account the effects of flexibility

on various physical properties of the system.

1.4.2 Flory Theory

The following discussion and derivation are reproduced from reference [8]. The Flory-Huggins theory uses some simple arguments to predict the correlations for various physical quantities for monomers in a polymer. For the freely-jointed chain ($\kappa = 0$), it predicts the end-to-end distance for a polymer with N monomers to be

$$R \propto bN^\nu \quad (1.6)$$

with $\nu = 1/2$ where b is the equilibrium bond-length, also known as the Kuhn length. The probability distribution for $\mathbf{R}$ is found to be

$$P(\mathbf{R}, N) \approx \left(\frac{3}{2\pi Nb^2} \right)^{3/2} \exp -\frac{3}{2} \frac{R^2}{Nb^2} \quad (1.7)$$

in three dimensions. This can then be used to find the entropy of the chain. As discussed before, we must also take into account excluded volume interactions to realistically model polymers. This contributes to the free energy of the chain, and this contribution is proportional to the excluded-volume for each monomer, and the number of possible monomer-monomer interactions (N^2), but inversely proportional to the amount of total available volume (R^3). Thus, the total free energy of the chain is given by

$$F_N(R) = F_0 + e_0 \frac{R^2}{Nb^2} + e_1 v_{\text{exc}} \frac{N^2}{R^3} \quad (1.8)$$

where F_0 represents any other contributions to the free-energy (which are considered to be independent of R), and e_0, e_1 are proportionality constants. In order to minimize the free energy with respect to R , the two R -dependent terms must be

equated, which gives us

$$R \propto N^\nu b^{1-\nu} \quad (1.9)$$

with $\nu = 3/5$ in three dimensions. Generalising to d dimensions, $\nu = 3/(d + 2)$. This quantity is also known as the Flory radius.

These results were also used by Alexander and de-Gennes to successfully propose similar scaling laws for polymer brushes.

1.4.3 Grafted Polymer Brushes

Polymer brushes are systems with multiple polymers grafted onto a flat surface. One end of a polymer is usually considered to be fixed at the surface, and immovable, while the rest of the polymer is free to flail around due to thermal fluctuations, shear forces, or interactions with other polymers. The number of polymer chains per unit area is defined as the grafting density as shown in Eq. 1.10 (although, in some contexts, this may refer to a different quantity).

$$\sigma_g = \frac{N_{\text{chains}}}{A} \quad (1.10)$$

The physical properties of these brushes were successfully predicted by S. Alexander and P.G. de-Gennes, who laid the foundations of polymer brush theory. Their scaling laws for the height of a polymer brush as a function of its grafting density are described below.

1.4.4 Alexander-de-Gennes scaling laws

S. Alexander and P.G. de-Gennes laid the foundations for the scaling laws for polymer brushes, successfully predicting the relation between the average height of a polymer in a brush and the grafting density σ_g [9, 10]. In the low σ_g regime, or

more specifically, when $\sigma_g < a^{-2}N^{-6/5}$,¹ a being the effective segment length for the polymer, the polymers in the brush barely interact, and their average height is simply the Flory radius discussed earlier [9, 10]. This is called the ‘mushroom’ regime, as individual polymers resemble mushrooms with bulbous heads.

On the other hand, when $\sigma_g > a^{-2}N^{-6/5}$, the polymers are no longer independent and have significant interactions, causing the following scaling law to hold (Eq. 1.11) for the average height (also called the brush thickness) [9, 10].

$$L \approx Na(\sigma_g a^2)^{1/3} \quad (1.11)$$

These scaling laws may be derived in the following way. We consider each chain to be divided into correlated regions inside of which it behaves like a self-avoiding walk. The radius of these correlated blobs or spheres (d) would be the average lateral spacing between the polymer chains, which is the reciprocal of the square-root of the grafting density (σ_g) by definition.

$$d = \sigma_g^{-1/2} \quad (1.12)$$

Inside each blob, we assume that a chain behaves like a self-avoiding walk. Let the average number of monomers of the chain inside each blob be a constant g . Then, according to Flory theory, the radius d of each blob should be

$$\begin{aligned} d &\approx ag^{3/5} \\ g &\approx \left(\frac{d}{a}\right)^{5/3} \end{aligned} \quad (1.13)$$

where a is the segment length or Kuhn length as we have seen before. For a length with a total of N monomers, the number of blobs (n_b) is simply N/g , and the average

¹For the polymers to not interact at all, the spacing between them would have to be $2aN^{3/5}$, which corresponds to $\sigma = a^{-2}N^{-6/5}/4$. Even at 4 times this density, i.e. $\sigma = a^{-2}N^{-6/5}$, the interactions between the polymers are not very significant as they barely overlap.

total height of the chain is then

$$\begin{aligned}
L &\approx n_b d \approx \frac{N}{g} d \\
L &\approx \frac{N a^{5/3}}{d^{5/3}} d \\
L &\approx N a^{5/3} d^{-2/3} \\
L &\approx N a (\sigma_g^{1/3} a^2)
\end{aligned} \tag{1.14}$$

which is precisely the expression of the Alexander - de-Gennes scaling law.

In the beginning of this project, we verified these scaling laws for polymer brushes using molecular dynamics simulations in LAMMPS.

1.4.5 Virus - Glycocalyx interactions

As mentioned before, regulating the entry of viruses and other pathogens is an important function of the glycocalyx. In order to study such interactions between the pathogens and the glycocalyx, the pathogens are often modelled as nanoparticles that can be introduced into a polymer brush model for the glycocalyx. In this project, we also modelled the viruses as nanoparticles, and performed a literature review in order to choose appropriate interaction potentials for virus-virus and virus-chain interactions.

The most common choice for modelling viruses seems to be the Lennard-Jones potential, which we have already discussed before. A cutoff is usually introduced to limit long-range interactions and for computational feasibility. However, a small attractive well is usually preserved (unlike the WCA potential which is purely repulsive) in order to model the multivalent attraction to the polymer chains that many viruses are known to exhibit.

One of the main references for this project is a paper by Cui et al. [11], where

the authors use the Born-Mayer-Huggins (BMH) potential (eq. 1.15) to represent the virus-chain interactions. They chose this potential as it allows them to control the steric repulsion and the adhesive interactions independently by changing the independent parameters C and D .

$$E = A \cdot \exp\left(\frac{\sigma - r}{\rho}\right) - \frac{C}{r^6} + \frac{D}{r^8} \quad r < r_c \quad (1.15)$$

However, the minima of the BMH potential cannot be shifted significantly to alter the size of the nanoparticle or virus. Thus, they manually shift the distance r to be defined from the surface of the virus rather than the center, which is an arbitrary choice.

A different interaction potential that shows very similar behavior while avoiding this issue is the Morse potential (eq. 1.16). Here the minima of the potential can be shifted by simply changing r_0 , allowing for straightforward computation. The strength of the potential (both the repulsive and attractive parts) is controlled by D_0 , and α controls the width of the potential well and thus the effective range of the potential.

$$E = D_0 \left(e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)} \right) \quad (1.16)$$

We chose to use the Morse potential for this project due to the ease of computation it provides, and because it shows qualitatively similar behavior to the Lennard-Jones potential. A cutoff was introduced for computational efficiency, and the potential well was shifted accordingly to avoid a discontinuity.

The viruses must also bind to the cell-surface receptors after penetrating the glycocalyx. In order to simulate these interactions, we must first model the receptors themselves. Cui et al. [11] model these explicitly as very short polymers with the top-most bead interacting with viruses through a very strong binding potential. This ensures that once a virus reaches a surface receptor, it does not escape again.

However, in order to see significant binding, the number of surface receptors must be higher than even the number of longer polymer chains in the layer, which can introduce a huge computational cost.

Due to time constraints, and limited computing power, we chose to replace the bed of surface receptors with a flat homogeneous adsorption plane that would interact with only the viruses through a standard Lennard-Jones potential. The strength of this interaction was set to be significantly higher than all other interactions in the model to ensure that the viruses could not escape once they were adsorbed onto the surface. While this may alter the binding statistics slightly, this is not a significant issue as we are more interested in looking at the way the viruses enter the glycocalyx itself.

1.5 Project outline

In Chapter 2, we discuss the computational methods used throughout this project, and the procedure followed to obtain results.

While the aforementioned Alexander - de-Gennes scaling laws and Flory theory provide sufficient theoretical background for predicting the height of homogeneous polymer brushes, the applicability of these scaling laws to mixed brushes seems to not have been investigated in detail. One way to predict the height of a mixed brush may be to approximate the brush with a homogeneous brush with the same average polymer length. However, this was found to not be accurate, especially at higher grafting densities. In the first part of Chapter 3, we show this result obtained from mixed brush simulations, and also provide a possible theoretical explanation for the same.

We then introduce viruses modelled as nanoparticles of varying sizes in these systems, building on the work of Cui et al.[\[11\]](#) to study viral entry in mixed brushes, as

well as the effect of non-uniform attraction between the viruses and polymer chains of different lengths. The second part of Chapter 3 includes these results, along with a discussion based on the theory.

A discussion of the scope of the project, the results obtained, and the methodology, is provided in Chapter 4.

Chapter 2

Methodology

2.1 LAMMPS

Large-scale Atomic/Molecular Massively Parallel Simulator [12], or LAMMPS for short, is an open-source molecular dynamics package which has been used widely for simulating systems at the atomic and molecular scales. It is primarily designed for high-performance parallel computing and provides a library with a broad range of interaction potentials, boundary conditions, and integrators, making it suitable for simulating various systems in soft matter physics, including polymers and polymer brushes.

All of the simulations for this project were designed using LAMMPS (Stable version, 2nd August, 2023), and run on the Computer Science and Computational Biology High Performance Cluster at Ashoka University. A set of Python and Bash scripts were used to create the LAMMPS input files, data files, HPC job scripts, etc. The obtained data was post-processed and analysed using Python code.

2.1.1 Simulation setup

The details of the polymer brush simulations are discussed here.

The interactions between any two non-bonded monomers were modeled using the WCA potential discussed earlier [1.2](#) with a cutoff value of 1.12246 ($\approx 2^{1/6}$).

The bonds between two successive monomers were modeling with the FENE potential discussed earlier ([1.4](#)). In order to reduce the number of factors affecting the dynamics of the system, the proportionality constant k was set to thirty times ϵ , the maximum repulsion in the WCA potential. This allows us to treat the bonds as being virtually inextensible.

The polymers were grafted at the $z = 0$ plane in a square lattice, with the first monomer in each chain fixed at its grafted position throughout the simulation. A reflective plane was placed at $z = 0$, which prevents the polymers from passing through the grafting plane. A similar reflective plane was placed on top of the simulation box, but this was placed at a height where the polymers never interact with it.

In the X and Y directions, periodic boundary conditions were applied to remove boundary effects.

The lattice spacing is the parameter which is varied to obtain a desired grafting density, with the area of the box being scaled accordingly. To ensure that the polymers at the boundaries see a similar environment as those in the bulk, the periodic walls were always placed at a distance equal to half of the lattice spacing from the last rows of polymers in the X and Y directions. With the minimum image convention (which LAMMPS implements by default), this ensures that the boundary effects are removed, or at least minimised.

The viruses, when present, are modelled using the Morse potential as described

earlier. Equilibrium snapshots from simulations performed in the first part of the project were used as initial conditions for all viral-entry simulations, and the viruses were placed slightly above the average height of the longer polymers in the equilibrated brush. A reflective wall was added above the viruses to prevent them from diffusing away from the layer, which would otherwise cause most viruses to simply never enter the brush. This wall only interacted with the viruses, and had no effects on the polymer chains in the rare case that they extended so far as to reach it. The addition of such a reflective wall is slightly arbitrary but was necessary for computational feasibility. It is discussed in more detail in the Discussion chapter.

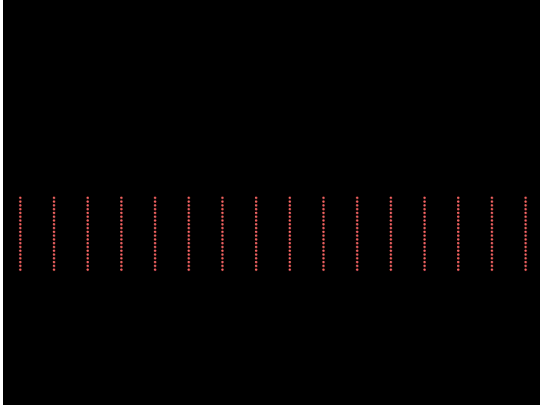
An adsorption plane was placed at $z=0$ to represent the bed of surface receptors as discussed previously. This plane also interacted only with the viruses, and had no interactions with the polymer chains in the brush.

The affinity of the virus-chain interactions was varied across a large range to observe its effect on the entry of viruses into the brush layer. The strength of the adsorption plane for the viruses was set to 5.0 LJ units, i.e., 50 times higher than the thermal fluctuations. This was done to ensure that viruses, once bound, would not escape the surface or diffuse into the brush again.

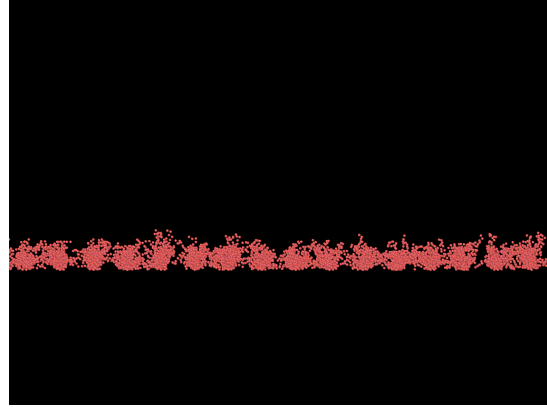
A Langevin thermostat was used, with a relaxation time τ that was always 100 times the time-step. This was used in conjunction with the NVE fix in LAMMPS, which simulates the canonical ensemble. A time-step of 0.006s was found to be sufficient for most simulations, provided the temperature around 0.1K.

A table with the chosen values of all relevant parameters in the simulations is provided in the Appendix.

Visualisations from the different kinds of simulations that were performed were created using Ovito, and are shown below.



The initial configuration of a homogeneous polymer brush.



A snapshot from the same brush taken at equilibrium.

Figure 2.1: Snapshots from one homogeneous polymer brush simulation. The bonds between successive monomers are not shown here for clarity.

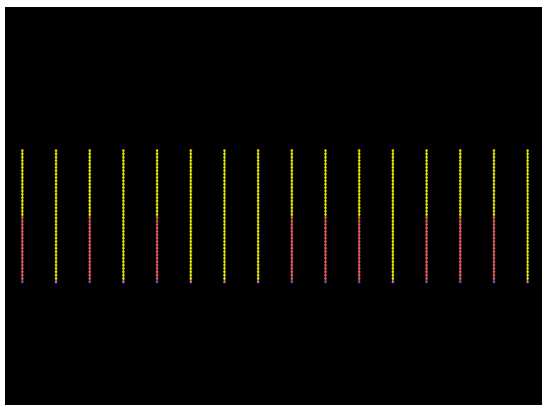
2.1.2 Workflow

A python script was used for creating input files which could be submitted as LAMMPS jobs on the HPC server. The script has the capability to produce simulations with homogeneous brushes, mixed brushes, viruses interacting with a non-equilibrated brush, and viruses interacting with an equilibrated brush where an equilibrium snapshot from a previous simulation is provided. All relevant parameters are accepted as inputs, and can thus be changed as required, with very few parameters being hard-coded into the script.

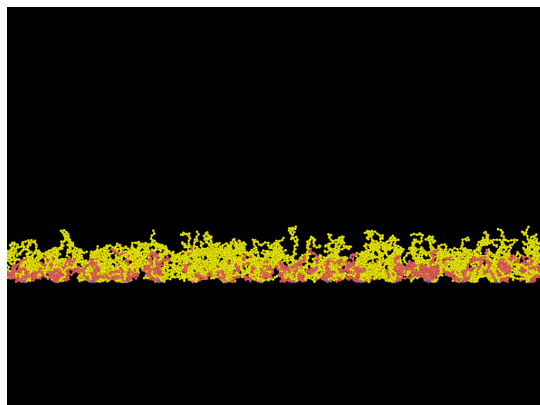
Bash scripts were used on the HPC server to create a large number of input files, or multiple seeds for each set of inputs, and to create, schedule and run the jobs. A combination of bash and python scripts was again used to analyse the output from the simulations, and to obtain the desired results.

All scripts used for this project can be found in the following github repository : <https://github.com/eft-isimus/ug-thesis>.

All visualisations of the simulations for verification, as well as for presenting the results, were created using Ovito.

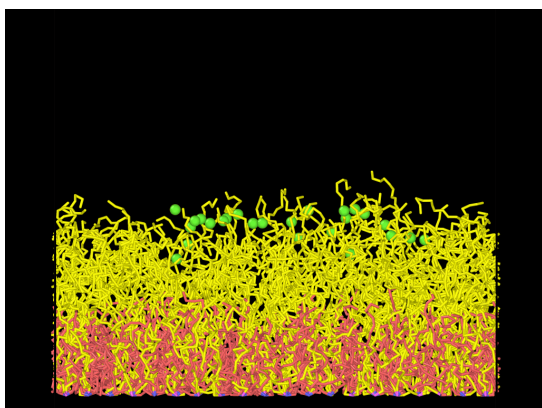


The initial configuration of a mixed polymer brush.

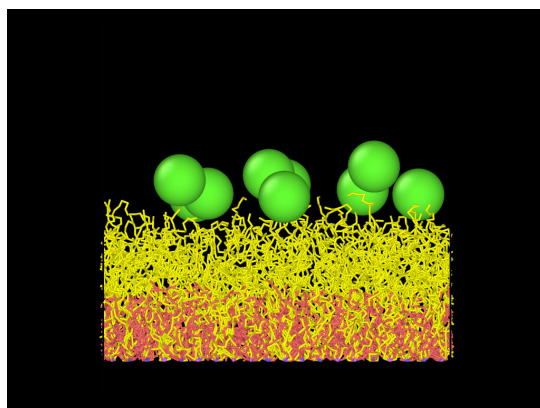


A snapshot from the same brush taken at equilibrium.

Figure 2.2: Snapshots from one mixed bi-disperse polymer brush simulation. The bonds between successive monomers are not shown here for clarity. The longer polymers are shown in yellow, while the shorter polymers are shown in red.

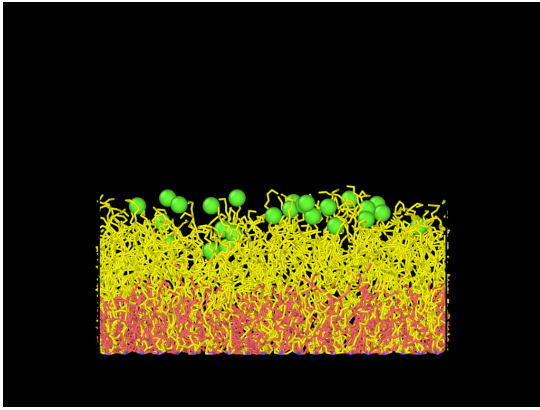


Virus radius = 0.625, the smallest size of viruses that were tested.

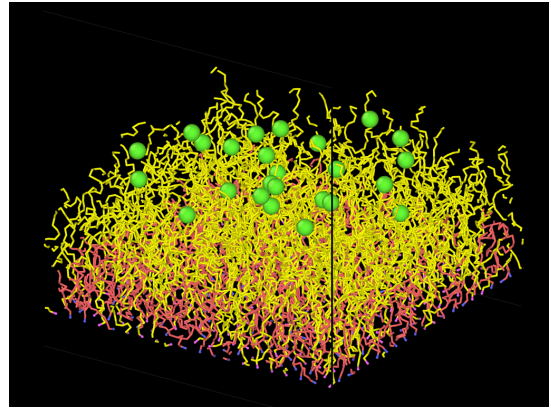


Virus radius = 3.75, the largest size of viruses that were tested.

Figure 2.3: Snapshots from some simulations showing the smallest and largest virus sizes introduced over an equilibrated mixed polymer brush. The longer polymers are shown in yellow, the shorter polymers in red, and the viruses in green.



A snapshot showing a front view of a simulation with viruses on top of a mixed brush



A snapshot showing a different view from the same simulation.

Figure 2.4: Snapshots from some simulations showing viruses of radius 1.25 introduced over an equilibrated mixed polymer brush. These were used to study the dependence of viral entry probability on the virus-polymer affinity. The longer polymers are shown in yellow, the shorter polymers in red, and the viruses in green.

Chapter 3

Results

A substantial portion of the first part of the project involved establishing a reliable simulation workflow on the HPC system and validating the LAMMPS simulation setup. Once this computational framework was in place, we began by verifying the Alexander–de Gennes scaling laws for flexible polymer brushes.

3.1 Verifying Alexander - de-Gennes scaling laws

To this end, simulations were performed for systems consisting of 400 polymer chains arranged on a square lattice. Each polymer had a length of 20 monomers and was grafted by one end to the $z=0$ plane. The simulations were carried out at a temperature of $T = 0.1$ in Lennard-Jones units, and the average height of the brush was calculated. The difference between the z -coordinate of the monomer that was farthest from the grafting plane, and the base monomer (which is always fixed at $z=0$), was taken to be the height of an individual polymer.

For each grafting density, results were averaged over 10 independent random seeds. We observed that the behavior of the brush can be divided into two distinct regimes: the Flory regime for low grafting densities, i.e. $\sigma_g \leq a^{-2}N^{-6/5}$, where the average

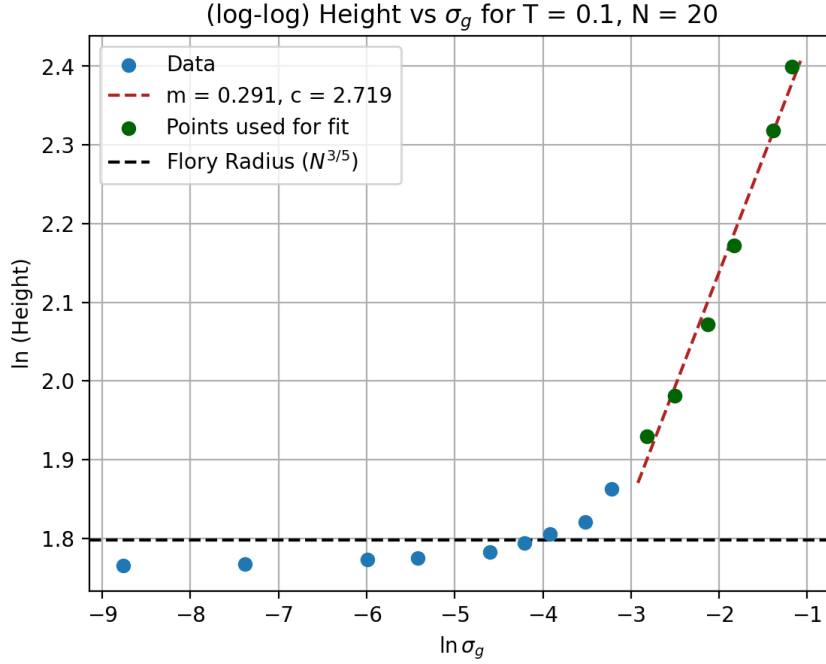


Figure 3.1: The equilibrium length at low grafting densities is found to be comparable to the Flory radius. At higher densities, the brush height is seen to scale with the grafting density.

height of the brush does not change significantly with an increase in grafting density, and the brush regime at higher densities, i.e. $\sigma_g \leq a^{-2}N^{-6/5}$, where the average height is seen to change significantly with a change in the grafting density. This is the region that was used to obtain a linear fit in the log-scale in order to obtain the scaling exponent.

3.1.1 Flory regime and brush regime

The preliminary result obtained with a handful of simulations as a proof of concept is shown in Figure 3.1. Note that the scaling exponent obtained here was significantly lower due to the lack of enough samples at higher grafting densities. This result is shown only to illustrate the Flory and brush regimes, and the value of the scaling exponent was improved significantly in later results, as shown in the following pages.

3.2 Studying mixed brushes

Having verified that our simulations obeyed the expected scaling laws, we conducted a large number of simulations for homogeneous brushes with polymers of length 20, 30, and 40 monomers (henceforth referred to as 20N, 30N, and 40N polymers respectively). We then moved on to simulating mixed bi-disperse brushes with polymers of two different lengths.

3.2.1 Mixed brushes follow scaling laws

We conducted simulations for mixed brushes with 20N and 40N polymers in varying proportions. The first mixed brush contained 20N and 40N polymers in an equal proportion (200 of each), which were randomly distributed on the grid. The average polymer length for this brush was thus 30N.

The second mixed brush contained a 9:1 ratio of 20N and 40N polymers distributed randomly on the grid, giving us an average polymer length of 22N.

The two distinct regimes depending on the grafting density were once again observed across all of these simulations, as shown in Figure 3.2. The data points above the critical grafting density were used to obtain the value for the scaling exponent. We observed that the height of mixed brushes also follows the Alexander - de Gennes scaling laws, with the same scaling exponent. We will return to a theoretical explanation for this in the following section.

We also noted that while the average length of polymers in the 50-50 mixed brush was 30N, the average height of the brush at all densities was lower than that of the homogeneous 30N brush (the blue and purple curves in Figure 3.3). This is an important result, and we discuss it further in the next section.

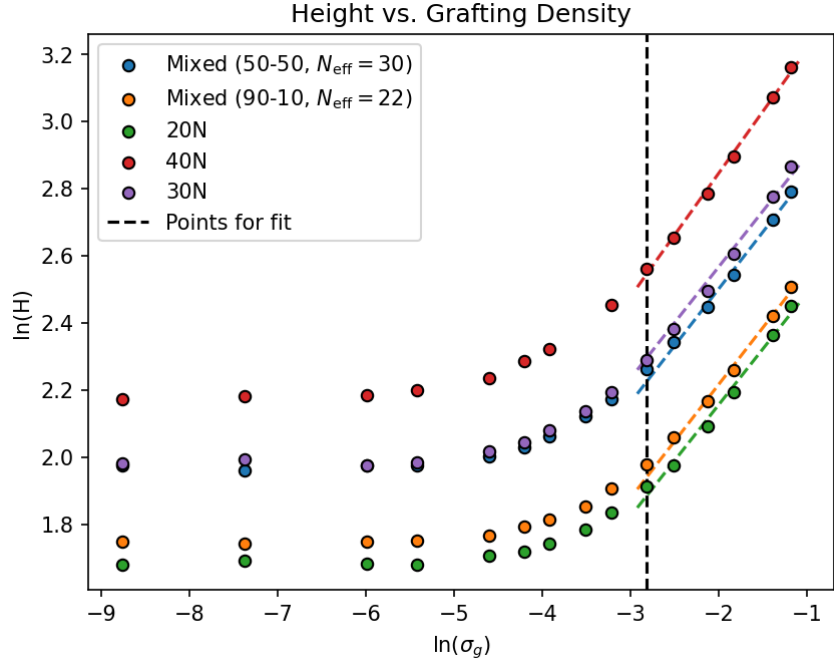


Figure 3.2: The behavior of all mixed as well as homogeneous brushes can be divided into the Flory and brush regimes. Only the data points above the critical grafting density should be used to calculate the scaling exponent for the brush regime.

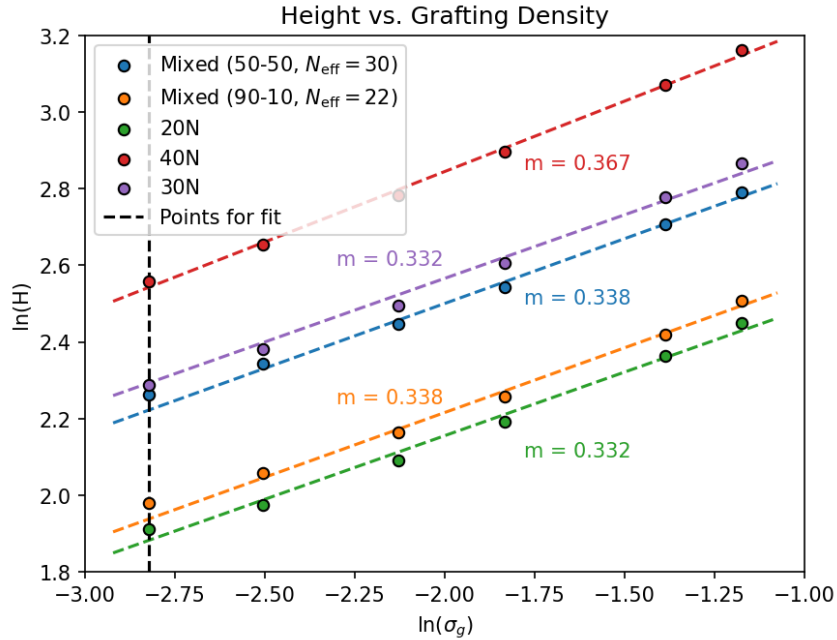


Figure 3.3: The scaling exponent was calculated for all mixed as well as homogeneous brushes. The value of the exponent obtained for each brush is shown next to the corresponding curve in the same color. The values obtained showed very good agreement with the value predicted by Alexander and de-Gennes (0.33).

3.2.2 Mixed brushes show unexpected behavior

As shown in Figure 3.3, the average height of a mixed brush is lower than the height of a homogeneous brush with the same average polymer length at all grafting densities in the brush regime. This is an important result because it means that we can not approximate a mixed brush with a homogeneous brush of the same average polymer length.

We propose that the reason for this discrepancy is as follows. We can divide the mixed brush into two regions, above and below the average height of the shorter polymers. The lower region contains all of the shorter polymers, as well as the lower segments of the longer polymers in the layer. All of these are grafted at the chosen grafting density, σ_g . However, above the shorter polymers, the upper segments of the longer polymers experience a less crowded environment, and thus a lower effective grafting density.

If they were to experience the same grafting density, the average height of the mixed brush would be equal to the height of the homogeneous brush. But since the upper segments of the longer polymers experience a lower density, they can equilibrate to a lower height, leading to a lower average height for the brush, which explains the discrepancy we observed.

We thus propose that the average equilibrium height of a mixed brush can be predicted by considering it to be composed of two brushes, one of length N_S (the number of monomers in the shorter polymers) that forms the base, and another with length $N_L - N_S$ (difference in the number of monomers between the longer and shorter polymers) that sits on top of the first brush and experiences a lower grafting density $p\sigma_g$, where p is the proportion of longer polymers in the brush. At equilibrium, the lower region can be considered to form a fixed base, much like the grafting plane at the very bottom.

We thus propose the following formula for predicting the average equilibrium height of a mixed brush (3.1). Here, a is the segment length or Kuhn length.

$$H = a^{5/3} N_S (\sigma_g)^{1/3} + a^{5/3} (N_L - N_S) (\sigma_g p)^{1/3} \quad (3.1)$$

We find that while this formula gives us a height lower than that predicted by replacing the mixed brush with a homogeneous brush of the same average polymer length, it still does not give us a height that matches the value obtained from simulations. The observed height is found to be slightly higher than the one predicted by this formula, which is to be expected for the following reason.

In proposing this formula, we assumed that the lower region, once equilibrated, acts like a fixed bed. We then assumed that the upper segments of the lower polymers can be considered to be a separate polymer brush grafted on a homogeneous fixed plane. However, the lower bed is not completely fixed, as the polymer fluctuate in height significantly. These fluctuations may randomly 'kick' the polymers in the upper layer, preventing them from equilibrating to their lowest height (i.e. the height they would equilibrate to if they were grafted on a fixed static plane). Due to time constraints, this hypothesis could not be verified, and we would like to study this in further detail in the future.

3.2.3 Height difference increases with grafting density

In order to better characterise the height difference between the mixed brush and the homogeneous brush, we also varied the proportion of longer and shorter polymers in the mixed brushes. For each value of the ratio between the number of shorter to longer polymers, we defined an 'effective' polymer length, $N_{\text{eff}} = N_S + pN_L$. The height of the mixed brush can then be compared directly to the height of a homogeneous brush with this effective polymer length.

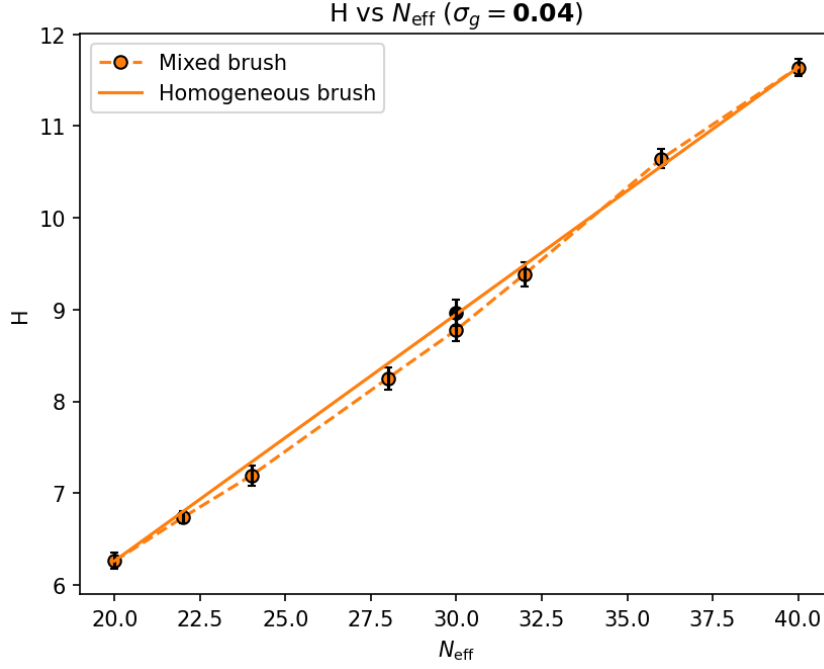


Figure 3.4: We see that the height of the mixed brush differs slightly from the homogeneous brush, but the difference is often between one error bar. The error bars here represent the standard deviation obtained from 5 seeds. The solid line shows the predicted value for a homogeneous brush based on the heights of the 20N, 30N, and 40N homogeneous brushes obtained from simulations.

We varied the ratio of the number of 20N polymers to 40N polymers from 10:0 (i.e., a homogeneous 20N brush) all the way to 0:10 (i.e., a homogeneous 40N brush), at a grafting density of $\sigma_g = 0.04$. We observed that the height of the mixed brush differed from the homogeneous brush slightly, and was usually lower, but the difference was not very significant, as shown in Figure 3.4

We then repeated this at a higher density, and found that the difference in height now becomes significant, and is no longer within the error bars, as shown in Figure 3.5. This tells us that the height difference increases with an increase in the grafting density, which is expected from the formula we proposed in equation 3.1. This is

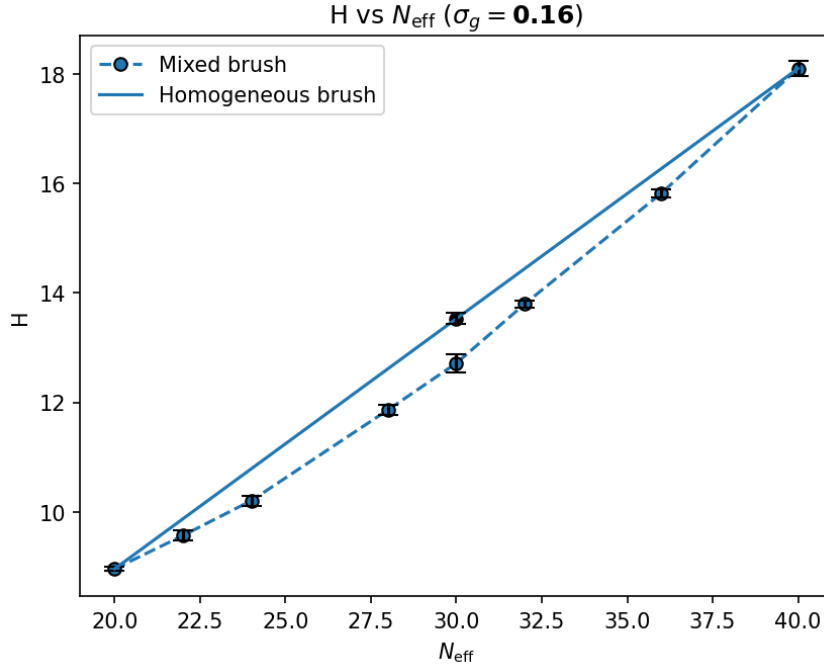


Figure 3.5: At a higher grafting density, we find that the difference in height becomes significant, and is no longer within the error bars. The difference is also found to be maximum at a 1:1 ratio of short to long polymers.

shown clearly in equation 3.2

$$\begin{aligned}
 H_{\text{diff}} &= H_{\text{hom.}} - H_{\text{mixed}} \\
 H_{\text{diff}} &= a^{5/3} \sigma_g^{1/3} (N_S(1-p) + pN_L) - a^{5/3} \sigma_g^{1/3} (N_S + (N_L - N_S)p^{1/3}) \quad (3.2) \\
 \therefore H_{\text{diff}} &\propto \sigma_g^{1/3}
 \end{aligned}$$

3.3 Studying viral entry into the glycocalyx

In order to study the entry of viruses into our polymer brush model of the glycocalyx, we introduced viruses over equilibrated snapshots of a mixed brush with 1:1 mixture of 20N and 40N polymers. As discussed before, the virus-polymer interactions were modelled using the Morse potential, and the affinities between the viruses and chains of each type could be controlled independently. The size of the viruses could also be controlled independently.

Based on the work by Cui et al. [11], we knew that with a multivalent attraction to the polymer chains, viruses can be engulfed into the brush layer. However, since we are using a different attraction potential for the virus-polymer interactions than the one used in the paper, as well as different units (Lennard-Jones units instead of SI units), we could not rely on the values they used for the interaction strength.

3.3.1 Size dependence

We began with the simpler situation where the viruses do not have any affinity for the chains in the brush. Equilibrium snapshots from a brush with a grafting density of 0.16 (the units being chains per unit length-squared) were used. The spacing between two adjacent polymers in this brush was 2.5 length units. The radius of the viruses was varied from 0.625 units to 3.75 units in order to observe how viruses of different sizes interact with the layer.

We simulated 16 viruses of the same size, which were introduced a few length units above an equilibrated mixed brush (the initial height was chosen depending on the size of the virus to ensure no overlap between the virus atoms and the polymer atoms, or to prevent an initial state with a very high potential energy). A reflective wall that could only interact with the viruses was placed above the viruses in order to prevent them from diffusing away from the brush, which would reduce the probability of the virus ever entering or even reaching the layer drastically. This means we are assuming the existence of an unexplained mechanism that drives the virus close enough to the brush for the multivalent attraction to come into effect, but what such a mechanism may be is beyond the scope of this project.

As mentioned earlier in the Introduction chapter, we also simulate a flat homogeneous adsorption plane below the brush with an adsorption strength that is a hundred times the thermal fluctuations. The plane only interacts with virus atoms,

and through a Lennard-Jones potential with a short cutoff. This ensures that the only role it plays in the simulations is that of preventing viruses that have reached the surface from diffusing away again.

In order to quantify the probability of a virus of a given size being able to enter the brush, we look at the probability density as a function of the virus height (or its z -coordinate) throughout the simulation. While we are not looking at the equilibrium probability distribution, as that would often require significantly longer (and thus more computationally expensive) simulations, this probability density over the entire simulation can still give us an idea of whether or not the viruses are able to enter the brush over a specific time-scale.

Figure 3.6 shows the probability density as a function of virus height for viruses of various radii. We see that the viruses of radius 0.625 length units and 1.25 length units are able to simply diffuse through the brush even in the absence of any affinity to the polymer chains. This is expected, since a virus of radius 0.625 (or diameter 1.25) can fit easily through the average expected gap between the polymer chains, which would be close to 2.5 length units (the polymer spacing for the brush). Similarly, a virus of radius 1.25 (or diameter 2.5) length units can fit exactly between two chains. Due to this, these viruses do not lead to a significant decrease in the conformational entropy of the polymer chains if they enter the layer, and are thus able to simply diffuse through the layer.

On the other hand, viruses of radius 2.5 (or diameter 5.0) length units, or 3.75 (or diameter 7.5) length units can not fit through the gaps between the polymer chains. Their introduction into the layer would thus drastically reduce the conformational entropy of the polymer chains by restricting the space available to them, and is thus not favored entropically. This entropic as well as steric barrier prevents these viruses from entering the layer, and they simply float on top of the brush in the absence of any virus-chain affinity. We then proceeded to study how having an affinity to the

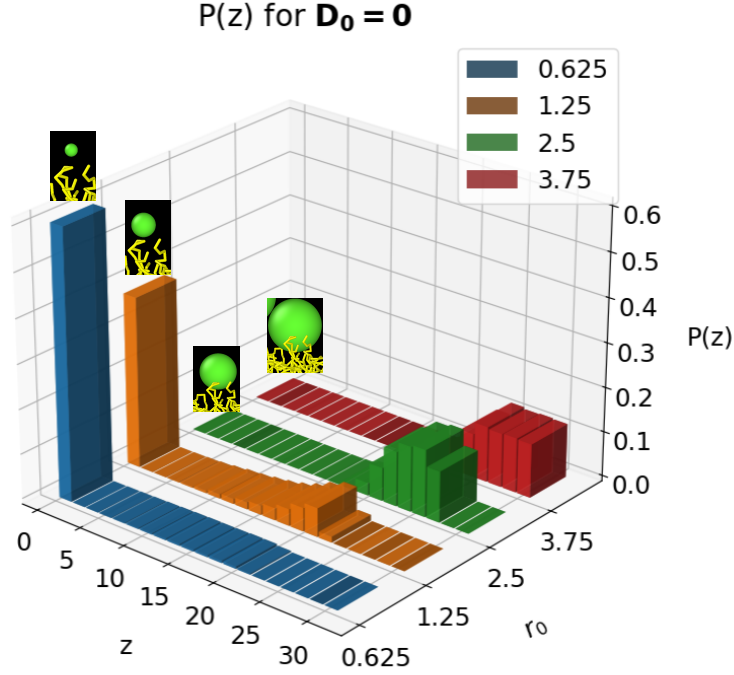


Figure 3.6: Viruses small enough to fit in the gap between the polymers diffuse through the brush. Larger viruses are arrested outside the brush.

polymer chains may help the viruses enter the layer.

3.3.2 Affinity dependence

We tested the effects of adding a virus-chain affinity using viruses of radii 1.25 and 2.5 length units. The affinity was first kept uniform for both types of chains, with its strength being varied from 0 to 2.5 energy units. Figure 3.7 shows the probability distributions for viruses of radius 1.25 with various affinity strengths.

We have already discussed how these viruses were able to diffuse through the layer without any affinity to the polymer chains. We further found that adding a small affinity of even 0.001 energy units increased the probability of these viruses reaching the surface to nearly 1. A higher affinity of up to 0.1 energy units still seemed to sustain this high entry probability. However, upon increasing the affinity even

further to 1.0 energy units, we found that the viruses are not always able to reach the surface and a second peak appears in the histogram around $z = 10$ or higher. This peak is found to shift further to the right with a higher affinity of 2.5 energy units. In order to observe this more clearly, we plot the probability densities for these two affinities only, along with markers to indicate the average height of the lower and upper layers of the brush, in Figure 3.7. We found that the viruses were able to enter the brush, but were getting stuck above the lower layer of polymers. At the highest affinity strength that we tested, 2.5 energy units, we see that the viruses are not able to enter the lower layer at all.

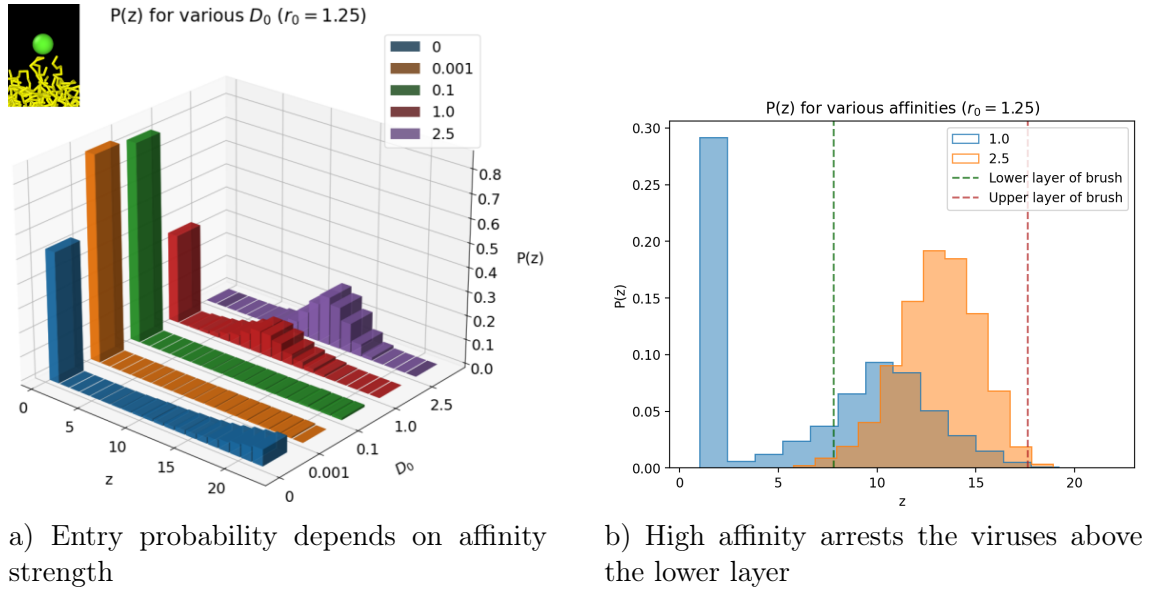


Figure 3.7: Low but non-zero affinity strengths allow the viruses to enter the brush, but higher affinities arrest them above the lower layer of the brush.

This can be verified from the results obtained for viruses of radius 2.5 units as well. The probability densities for various affinities for these viruses are shown in Figure 3.8. Unlike the smaller viruses, we find that these cannot enter the brush at all without any affinity, and the probability density curves are all peaked above or near the average height of the upper layer of the brush for small affinity strengths of up to 0.2 energy units. With an affinity strength of 0.5 energy units, we find that the probability density peak suddenly shifts to the other end, with the viruses having a

very high probability of entry.

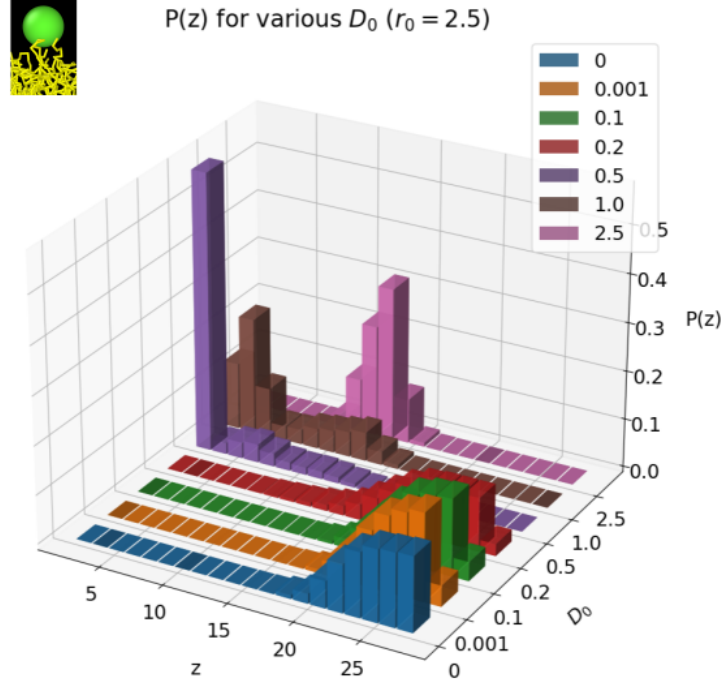


Figure 3.8: Viruses that are too large to simply diffuse through the brush require a critical affinity at which entry becomes possible.

However, increasing the affinity strength further again leads to the peak shifting right-ward. For an affinity of 1.0 energy units, the brushes have a high probability of being trapped above the lower layer, and at a strength of 2.5 energy units the probability density is again entirely within the outer layer of the brush.

Based on these results, we predict that for a given radius of the viruses, a given brush grafting density (or, equivalently, polymer spacing in the brush), and a given temperature, there may exist a critical affinity strength, below which the viruses cannot enter the brush at all, and above which they may enter the outer layer but not the lower layer of the brush.

Since virus engulfment involves a competition between the virus-polymer affinity and thermal fluctuations, a higher temperature would likely cause a higher corresponding critical affinity. Similarly, since the engulfment of larger viruses would lead to a larger decrease in the conformational entropy of the polymer chains, the critical

affinity would also increase with an increase in virus radius. Below a certain radius, the viruses can simply diffuse through the layer, and a critical affinity would thus not be observed.

A possible explanation for the viruses getting arrested above the lower layer of the brush may be as follows. A high affinity strength can overcome the decrease in conformational entropy of the polymers which would prevent the entry of the viruses into the brush. However, once the virus is engulfed in the upper layer of the brush, it may form many connections with various polymer atoms. In order to penetrate deeper into the brush, it would have to sever these connections and explore the denser layer, where it would then form an even larger number of connections, reaching a more stable state. However, the only mechanism that can lead to such detachment is the thermal fluctuations, which may not be strong enough to cause detachment at high affinity strengths. The viruses may thus reach a very stable state in the upper layer of the brush, with the thermal fluctuations not being strong enough to push them over into a deeper energy minima in the lower layer of the brush.

At lower affinities, even though the viruses may be in a stable state in the upper layer of the brush, the thermal fluctuations may be comparable in strength to these connections, allowing the viruses to explore the lower layer of the brush. Another way to ensure this happens would be to increase the affinity that the viruses have to the shorter polymer chains in the brush. In order to test this, we simulated viruses which have a fixed affinity to the longer polymers (denoted by D_0^L), and various values of affinity to the shorter polymers. The probability densities for this are shown in Figure 3.9

We found that as the shorter polymer affinity increases with respect to the longer polymer affinity, the viruses are able to penetrate deeper into the brush, as shown by the probability density peaks shifting towards the left. Having an equal but high

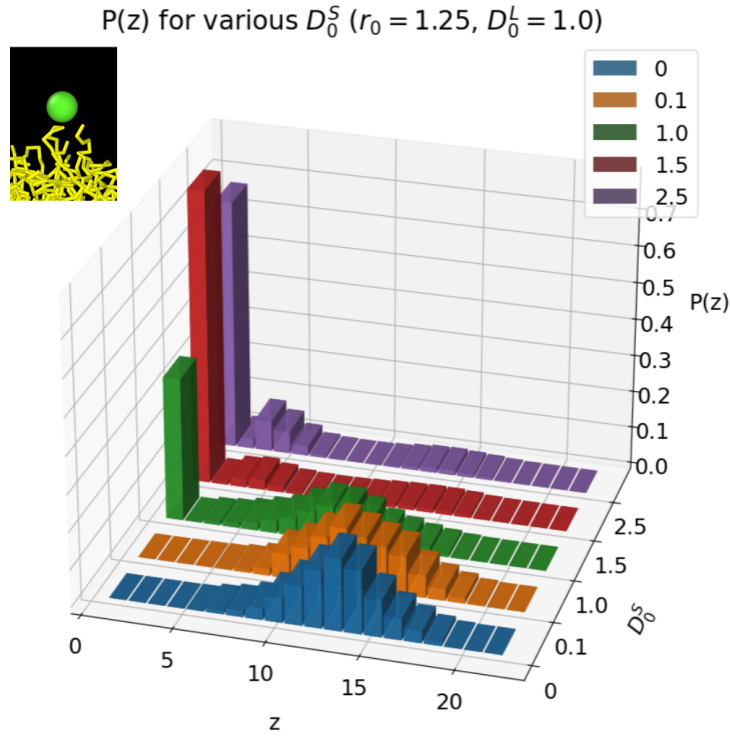


Figure 3.9: Viruses of radius 1.25 units are able to enter the lower layer of the brush if they have a higher affinity for shorter polymers than for the longer polymers.

affinity for both types of chains, like 1.0 energy units, the viruses are able to enter the brush entirely, but with only around a 40% probability. Finally, with a shorter polymer affinity that is higher than the longer polymer affinity, the viruses are able to enter the brush entirely with a high probability.

Chapter 4

Discussion

4.1 Restricting virus diffusion

As mentioned previously, we place a reflective, impenetrable wall above the viruses in order to study their entry into the brush. This prevents the viruses from simply diffusing away from the brush instead of entering it, which is always entropically favorable for the system. Depending on how far the viruses are placed from the equilibrated brush, they may take a very long time to reach the layer, and a significantly larger number of samples would be needed since a small proportion of the total simulated viruses would end up interacting with the brush. Thus, the impenetrable wall serves as a restriction on the system which allows us to explore the specific configurations that we are interested in, namely the ones where the virus is interacting meaningfully with the brush.

Another way to do so would be to place the viruses at the precise z -coordinate where they are grazing or touching the equilibrated brush, which would reduce the chance of them diffusing away in a similar fashion as the reflective wall. However, this may introduce instabilities in the system, since small adjustments in the initial positions of the viruses could lead to large repulsive forces, and thus a very high

initial potential energy.

This method would also require computing a suitable height for viruses of different radii. Additionally, since the polymers in the brush fluctuate around their average height, the viruses may escape even after placing them at the appropriate height, once again leading to a lot of samples that explore uninteresting configurations where the viruses simply diffuse away from the brush. Therefore, the method we used seems to be the most computationally feasible way to explore the configurations of interest, and is also the method of choice for Cui et al. whose paper on viral penetration in the Glycocalyx by viruses with weak multivalent adhesive attraction was the main reference for our work. The locomotion mechanisms that may help viruses reach the Glycocalyx in the first place are beyond the scope of this work, but may be of interest for future studies.

4.2 Approximating surface receptors

The cell surface is crowded with receptors that detect and bind with external ligands which may be signals from other cells or pathogens. Viruses have evolved to be able to bind with some of these receptors on host cells in order to infect them.

The entry of viruses through the Glycocalyx thus usually results in their binding with the cell surface receptors, and subsequently infecting the host cell. Once the virus is bound to the receptor, it is unlikely to sever this bond and diffuse into the Glycocalyx again. Thus, in our simulations, reaching the cell surface was required to be simulated as an irreversible process. Cui et al. achieve this by explicitly modelling the cell surface receptors as small polymers with individual binding beads that interact with the viruses. These receptors must also exceed the number of polymers in the layer by around 50%, adding a significant computational cost.

However, for a majority of the simulation time, before any viruses have reached

the surface, simulating the dynamics of these small polymers serve no purpose as they are highly unlikely to affect the dynamics of the longer brush polymers. This computational cost can thus be avoided by instead replacing the receptors with a homogeneous adsorption plane that interacts with only the virus atoms. By making the adsorption strength significantly higher than the thermal fluctuations in the system, we ensure that the viruses do not escape and diffuse into the brush again after reaching the cell surface.

In this work, we were interested only in the probability of viruses entering the Glycocalyx. If we were to study the probability with which they may bind with a receptor and thus cause infection, this approximation of cell surface receptors by an adsorption plane would not suffice, and the receptors would have to be modelled explicitly.

4.3 Equilibrium Sampling for viruses

To characterise the probability of viruses being able to enter the brush layer, we looked at the probability distribution in virus height across the entire duration of our simulations. Ideally, we should look at the equilibrium statistics of the simulation instead, which would also allow us to calculate the free-energy barrier posed by the brush to the entry of viruses. Some of the secondary peaks seen in the probability distributions we observed may disappear at equilibrium, as these viruses may not be stuck at a particular height in the brush but slowly diffusing towards the surface. However, it was found that this descent, especially at high values of the virus-polymer affinity, was very slow, and would require significantly longer and computationally expensive simulations.

With the sampling that we use, where we look at the probability density across the entire simulation instead, we may not be able to calculate the free energy barrier

posed by the brush, but we are still able to observe the effects of varying both the size of the viruses and the affinity they have to the polymer chains in the brush. However, in the future, we would like to verify the predictions we have made on the basis of such sampling by using equilibrium sampling instead.

Appendix

Table 4.1 shows the values or settings used for various important LAMMPS simulation parameters. For detailed explanations about each of the parameters, refer to the LAMMPS documentation found here: <https://docs.lammps.org/>.

Table 4.2 shows the range of values tested for various important parameters in the simulations, as well as the different interaction potentials used in all simulations. The justifications for these choices can be found in Chapter 2: Methods.

All of the original code written for this work can be found at <https://github.com/eftisimus/ug-thesis>.

Parameter	Value used	Parameter	Value used
Time step (dt)	0.006	Neighbor list style	bin
Temperature	0.1	Neighbor list rebuild delay	100
Langevin damping time	0.6 (100*dt)	Neighbor list build frequency	5
Neighbor list skin distance	1.0	Communication cutoff	3.0

Table 4.1: All parameter values given here are expressed in the relevant Lennard-Jones units.

Parameter	Minimum value	Maximum value
Polymer length	20N	40N
Virus radius	0.625	3.75
Polymer brush density	0.00015625	0.3086
Virus - polymer affinity strength	0	2.5

Interaction potentials		
Purpose	Potential used	Strength
Virus-polymer affinity	Morse potential	0 - 2.5
Virus-polymer self-exclusion	WCA potential	1.0
Virus-virus self-exclusion	WCA potential	1.0
Virus-surface adhesion	LJ potential	5.0
Polymer-polymer self-exclusion	WCA potential	1.0
Bonded monomer interactions	FENE	30.0

Table 4.2: All parameter values given here are expressed in the relevant Lennard-Jones units.

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